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Archimedes Smart Muscle and Smart Rope Underwater Boat-Vessel Surface-Contract Launch System

Volume 00 — Table of Contents and System Overview

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System Name: AMSR-UBSCLS

Archimedes Smart Muscle and Smart Rope Underwater Boat-Vessel Surface-Contract Launch System

Synthesizes CSMFAB000000000022 (Smart Rope Actuator) and CSMFAB000000000109 (Archimedes Spear Physics) into a multi-vessel kinetic launch platform using Aegis Telluric-Proof dielectric materials throughout.

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Why Underwater Origin?

Water is 816x denser than air at sea level. This creates three mechanical advantages:

1. **Structural reaction mass:** The screw pushes against water (not vacuum) — thrust per RPM is proportional to fluid density.
2. **Natural launch tube:** The water column constrains yaw instability during acceleration, acting as a guided barrel.
3. **Smart Rope surface contract geometry:** The rope contraction force delivers maximum impulse precisely where the spear exits maximum-density fluid — at the water-air interface.

The energy cascade:

Stage 1 (depth d to surface): Archimedean screw drive - Exit velocity at surface: v_1 = sqrt(2 * T_net * d / m_spear)

Stage 2 (surface contract, t < 50 ms): Smart Rope impulse - Additional velocity: delta_v = n * F_rope * t_contract / m_spear

Stage 3 (above surface): Gyrostabilized ballistic coast + optional second stage

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Performance Summary Table

Scale	Spear Mass	Rope Energy	Exit Velocity	Apogee	Orbital Stage
Nano	50 g	2.4 kJ	310 m/s	4.9 km	None
Micro	1 kg	18 kJ	600 m/s	18.4 km	None
Small	10 kg	95 kJ	1,380 m/s	97 km	Small SRM
Medium	100 kg	450 kJ	950 m/s	46 km	Medium SRM
Large	1,000 kg	1,800 kJ	600 m/s	18 km	Major SRM

Citations (Vol 00)

- CSMFAB000000000022 V2.0, Smart Rope Actuator, CSM 2026
- CSMFAB000000000040 V2.0, Smart Rope Production, CSM 2026
- CSMFAB000000000109 V2.0, Archimedes Screw Physics, CSM 2026
- CSMFAB000000000001 V2.0, Aegis-C Composite Shielding, CSM 2026
- Munson et al., Fluid Mechanics, 9th Ed, Wiley 2020
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